

Learning Objective 3.1: I can describe aperture distributions and calculate aperture efficiency for a given illumination.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **The uniform aperture.** The feed of a rectangular horn makes the field across its 24 cm broad dimension essentially constant in amplitude and phase. It operates at 12 GHz.

- (a) In one or two sentences, state the relationship between the aperture distribution and the far-field pattern, and say what the space frequency $u = (L/\lambda)\sin(\theta)$ does for you that the angle θ does not.

The far field is the Fourier transform of the aperture distribution: the space factor is the integral of the aperture field weighted by the phase $e^{jkx\sin(\theta)}$ across the opening. Written against u , the pattern shape depends only on the *shape* of the illumination, so one curve serves every aperture with that illumination; the size L/λ only sets how many degrees correspond to each unit of u .

- (b) Find the aperture length in wavelengths and the half-power beamwidth in the broad-dimension plane.

$$\lambda = \frac{3 \times 10^8}{12 \times 10^9} = 0.025 \text{ m} = 2.5 \text{ cm}$$

$$\frac{L}{\lambda} = \frac{24}{2.5} = 9.6$$

$$\theta_{\text{HP}} = 50.8^\circ \frac{\lambda}{L} = \frac{50.8^\circ}{9.6} = 5.29^\circ$$

$$\theta_{\text{HP}} = 5.29^\circ$$

- (c) Find the angle of the first null measured from broadside.

The sinc pattern nulls at $u = 1$, so

$$\sin(\theta) = \frac{\lambda}{L} = \frac{1}{9.6} = 0.1042$$

$$\theta = 5.98^\circ$$

$$\theta_{\text{null}} = \pm 5.98^\circ$$

- (d) State the first sidelobe level of this aperture, and explain in one sentence why lengthening the horn would not change it.

The first sidelobe of a uniformly illuminated aperture is -13.3 dB relative to the peak. The normalized pattern $\sin(\pi u)/(\pi u)$ contains no L at all, so lengthening the aperture rescales the angle axis (narrower beam, higher gain) without changing the height of any lobe. The shape of the illumination sets the sidelobe level, and the uniform shape is fixed at -13.3 dB regardless of size.

2. **Aperture efficiency.** For a low-loss aperture antenna ($\eta_{\text{rad}} \approx 1$), aperture efficiency converts physical area into gain:

$$G = \eta_{\text{ap}} \frac{4\pi A}{\lambda^2}$$

- (a) Write the definition of η_{ap} in terms of the aperture field, and state in words what the numerator and the denominator each represent.

$$\eta_{\text{ap}} = \frac{|\int E_a dS'|^2}{A \int |E_a|^2 dS'}$$

The numerator is the coherent sum of the aperture field, which is what arrives on boresight when every point adds in phase. The denominator is the area times the power the aperture radiates. The ratio is coherent gain over available gain, it never exceeds one, and it reaches one only for constant amplitude and constant phase.

- (b) A line aperture carries a triangular illumination, $E_a = 1 - 2|x|/L$. Evaluate the two integrals over $\xi = x/L$ and compute η_{ap} .

$$\eta_{\text{ap}} = \boxed{0.750}$$

$$\int_{-1/2}^{1/2} (1 - 2|\xi|) d\xi = \frac{1}{2}$$

$$\int_{-1/2}^{1/2} (1 - 2|\xi|)^2 d\xi = \frac{1}{3}$$

$$\eta_{\text{ap}} = \frac{(1/2)^2}{1 \times (1/3)} = 0.750$$

This is the canonical triangular value, a gain penalty of $10 \log_{10}(0.75) = -1.25$ dB against uniform.

- (c) A pyramidal horn has an 8 cm $\times$ 6 cm mouth and a measured aperture efficiency of 0.51 at 10 GHz. Find its gain in dBi.

$$G = \boxed{15.3 \text{ dBi}}$$

$$\lambda = 0.03 \text{ m}, \quad A = 0.08 \times 0.06 = 0.0048 \text{ m}^2$$

$$\frac{4\pi A}{\lambda^2} = \frac{4\pi(0.0048)}{(0.03)^2} = 67.0$$

$$G = 0.51 \times 67.0 = 34.2$$

$$G_{\text{dBi}} = 10 \log_{10}(34.2) = 15.3 \text{ dBi}$$

- (d) Find the effective aperture A_e of that horn, and say in one sentence what the difference between A_e and the mouth area physically represents.

 $A_e =$ 0.00245 m^2

$$\begin{aligned} A_e &= \eta_{ap} A = 0.51 \times 0.0048 \\ &= 0.00245 \text{ m}^2 = 24.5 \text{ cm}^2 \end{aligned}$$

The missing 23.5 cm^2 is area whose field still radiates but does not contribute coherently on boresight, mostly because the horn's flare puts a quadratic phase error across the mouth and because the field tapers toward the edges. The horn dissipates none of it as heat, which is why η_{ap} is not radiation efficiency.

3. **The taper trade.** A line aperture is 40λ long. The table below gives the canonical taper values.

Illumination	First sidelobe	HPBW ($\times \lambda/L$)	η_t
Uniform	−13.3 dB	0.886	1.00
Cosine	−23 dB	1.19	0.81
Triangular	−26.5 dB	1.28	0.75
Cosine ²	−31.5 dB	1.44	0.667

(a) Find the half-power beamwidth with uniform illumination.

$$\theta_{\text{HP}} = \frac{50.8^\circ}{40} = 1.27^\circ$$

$$\theta_{\text{HP}} = 1.27^\circ$$

(b) The system requires a first sidelobe no higher than −25 dB. Choose the illumination from the table that meets it with the narrowest beam, and find the beamwidth you get.

Cosine gives only −23 dB, so the narrowest-beam choice that meets −25 dB is triangular at −26.5 dB :

$$\begin{aligned}\theta_{\text{HP}} &= 1.28 \frac{\lambda}{L} = \frac{1.28 \times 57.3^\circ}{40} \\ &= 1.83^\circ\end{aligned}$$

$$\theta_{\text{HP}} = 1.83^\circ$$

(c) Find the gain penalty in dB of this taper relative to uniform illumination on the same aperture.

$$\Delta G = 10 \log_{10}(0.75) = -1.25 \text{ dB}$$

$$\Delta G = -1.25 \text{ dB}$$

(d) Suppose the uniform beamwidth of part (a) is also a requirement. How long must the triangular-illuminated aperture be, and how much longer is it than the uniform aperture with the same beamwidth? Answer with a number and one sentence.

$$L' = 40\lambda \times \frac{1.28}{0.886} = 57.8\lambda$$

$$L' = 57.8\lambda$$

The aperture must be 44% longer to hold the same beamwidth. Lower sidelobes require either a wider beam or a longer aperture, and on a real platform the length is usually the binding constraint.

4. **Scaling a reflector.** A circular reflector of diameter $D = 1.2$ m is lit nearly uniformly by a broad feed, so much of the feed's power spills past the rim; its measured aperture efficiency is 0.60 at 10 GHz. Use the uniform circular aperture of Lesson 14:

$$\theta_{\text{HP}} = 1.029 \frac{\lambda}{D}.$$

- (a) Find the half-power beamwidth at 10 GHz.

$$\begin{aligned}\lambda &= 0.03 \text{ m} \\ \theta_{\text{HP}} &= 1.029 \frac{0.03}{1.2} = 0.0257 \text{ rad} = 1.47^\circ\end{aligned}$$

$$\theta_{\text{HP}} = \boxed{1.47^\circ}$$

- (b) Find the gain in dBi at 10 GHz.

$$\begin{aligned}A &= \pi \left(\frac{1.2}{2} \right)^2 = 1.131 \text{ m}^2 \\ \frac{4\pi A}{\lambda^2} &= \frac{4\pi(1.131)}{(0.03)^2} = 1.579 \times 10^4 \\ G &= 0.60 (1.579 \times 10^4) = 9475 \\ G_{\text{dBi}} &= 10 \log_{10} (9475) = 39.8 \text{ dBi}\end{aligned}$$

$$G = \boxed{39.8 \text{ dBi}}$$

- (c) The same dish operates at 30 GHz with the same aperture efficiency. Find the new beamwidth and the new gain.

Tripling the frequency triples D/λ :

$$\begin{aligned}\theta_{\text{HP}} &= \frac{1.47^\circ}{3} = 0.491^\circ \\ \Delta G &= 20 \log_{10} (3) = 9.5 \text{ dB} \\ G_{\text{dBi}} &= 39.8 + 9.5 = 49.3 \text{ dBi}\end{aligned}$$

$$\theta_{\text{HP}} = \boxed{0.491^\circ}$$

$$G = \boxed{49.3 \text{ dBi}}$$

- (d) The dish did not change, yet the gain rose by 9.5 dB. Explain where that gain came from, and name one practical reason a real reflector will fall short of the predicted 49.3 dBi at 30 GHz.

Gain is

$$G = \eta_{\text{ap}} \frac{4\pi A}{\lambda^2}$$

so it depends on area measured in square wavelengths. At 30 GHz the same 1.131 m^2 spans nine times as many square wavelengths, which is a factor of nine or 9.5 dB. In practice the aperture efficiency will not hold. The surface error σ is fixed in meters, and Lesson 14's Ruze loss is $685.8(\sigma/\lambda)^2$ dB, so tripling the frequency multiplies it by nine: $\sigma = 0.6$ mm costs 0.27 dB at 10 GHz and 2.5 dB at 30 GHz.

5. **Reading a measured pattern.** A student measures the pattern of an unknown aperture antenna at 10 GHz on the course range. The main lobe is symmetric about boresight, the first nulls fall at $\pm 3.4^\circ$, and the first sidelobes read 19 dB below the peak.

- (a) Assuming the illumination is uniform, find the aperture dimension in the measured plane.

$$\begin{aligned}\sin(\theta_{\text{null}}) &= \frac{\lambda}{L} \\ L &= \frac{0.03}{\sin(3.4^\circ)} = \frac{0.03}{0.0593} \\ &= 0.506 \text{ m} = 16.9\lambda\end{aligned}$$

$$L = 0.506 \text{ m}$$

- (b) Predict the half-power beamwidth that this aperture should show.

$$\begin{aligned}\theta_{\text{HP}} &= 0.886 \frac{\lambda}{L} = 0.886 \frac{0.03}{0.506} \\ &= 0.0525 \text{ rad} = 3.01^\circ\end{aligned}$$

$$\theta_{\text{HP}} = 3.01^\circ$$

- (c) The measured sidelobe is 19 dB down, not 13.3 dB down. Give two distinct explanations, one about the antenna and one about the measurement.

The antenna explanation is that the illumination is not uniform. A sidelobe between -13.3 dB and -23 dB is what a mild amplitude taper produces, somewhere between uniform and cosine. A taper moves the first null out to about $u = 1.2$, so the measured nulls imply a longer aperture, about 0.61 m, and a beamwidth near 2.8° , narrower than part (b) predicted. The measurement explanation is that the sidelobe was not resolved: too coarse an angular step, a range shorter than $2D^2/\lambda$, or reflections from the room all fill in nulls and can raise or lower a narrow sidelobe by several dB.

- (d) The same antenna is later found to have $\eta_{\text{ap}} = 0.44$, although its amplitude taper alone would predict about 0.95. Name two mechanisms that account for the difference, and state whether each one belongs to aperture efficiency or to radiation efficiency.

Phase error across the aperture and spillover past the aperture edge both belong to aperture efficiency: the power leaves the antenna, but it does not add coherently on boresight. Feed and strut blockage and cross-polarization are two more of the same kind. Conductor and dielectric loss in the feed structure belongs to radiation efficiency instead, because that power becomes heat and never radiates at all. Measured gain carries both, which is why $G = \eta_{\text{rad}} D$ and $D = \eta_{\text{ap}} 4\pi A/\lambda^2$ are separate statements.

Documentation: _____